

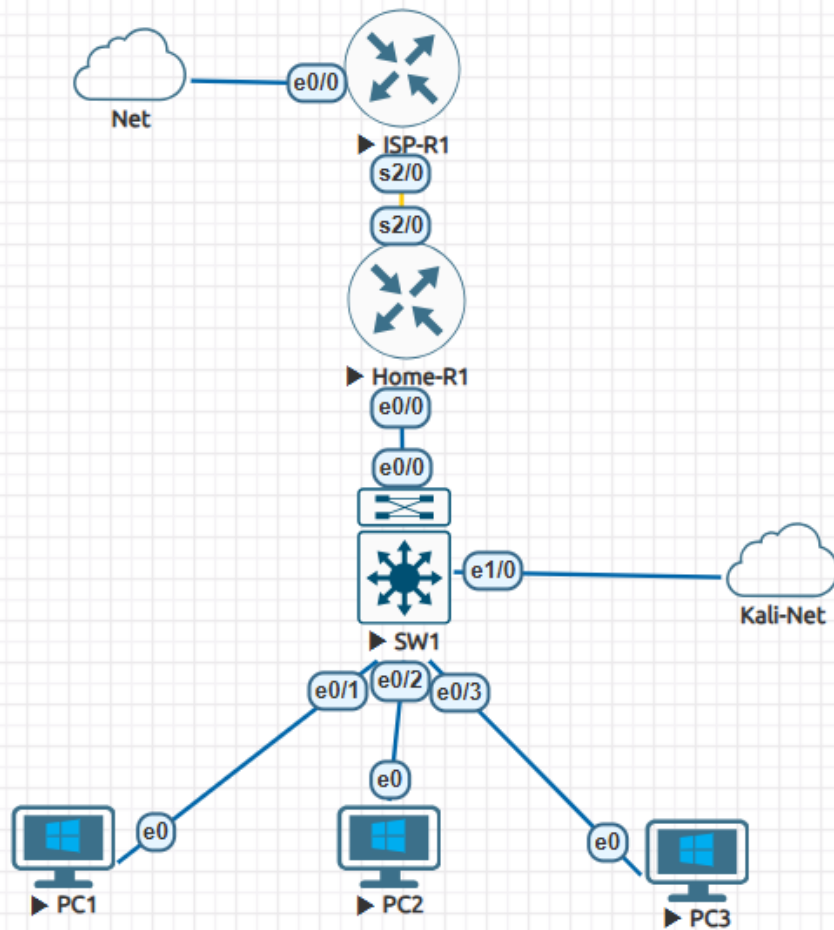
# PROJECT 3

## NOC Network Monitoring & Incident Response Lab Using Zabbix and EVE-NG

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Role: NOC / SOC Analyst (Simulated Lab Environment)

Date: 07/28/26



# 1. Project Objective

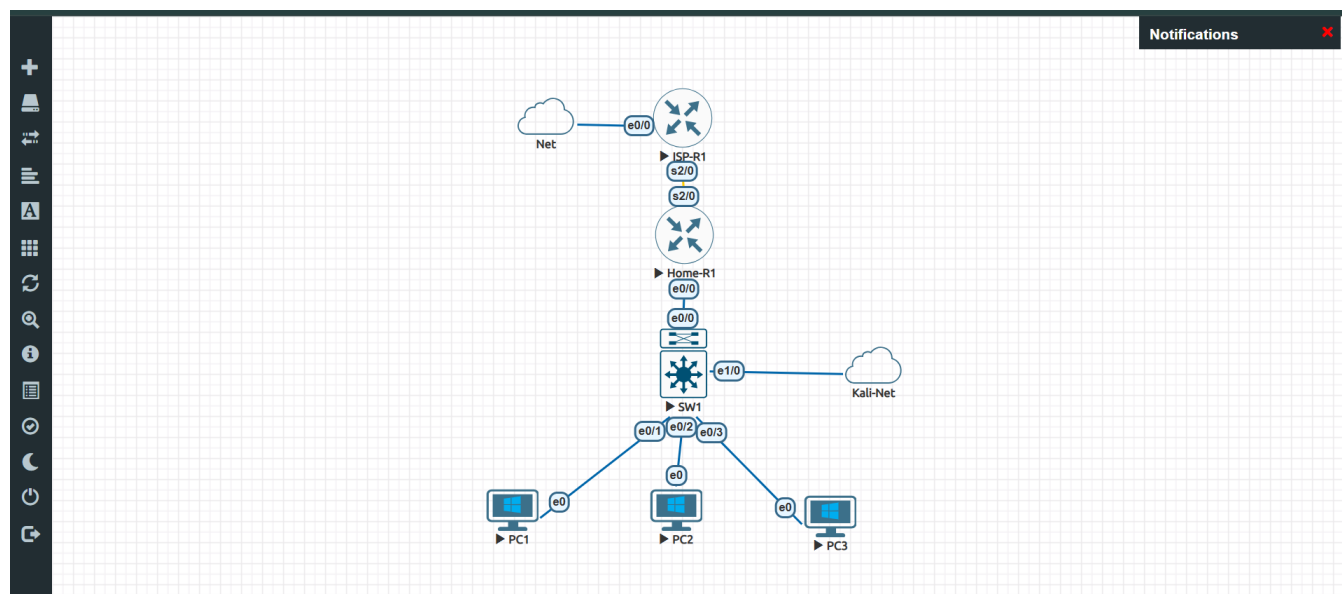
This project demonstrates the design and implementation of a simulated Network Operations Center (NOC) monitoring environment. Zabbix, running on Kali Linux, was used to monitor core network infrastructure devices (ISP-R1, Home-R1, and SW1) using ICMP and SNMP monitoring. The project covers host configuration, item and trigger creation, alert generation, and a full incident response lifecycle from detection through investigation, resolution, verification, and documentation.

**This project demonstrates the following skills:**

- Network monitoring configuration (ICMP and SNMP)
- Zabbix host, item, and trigger configuration
- Alert and problem management
- Incident investigation and root cause analysis
- Incident resolution and verification
- Professional incident documentation

# 2. Network Topology

The lab topology consists of an ISP router (ISP-R1) connected to a home router (Home-R1), which connects to a switch (SW1) serving three end-user PCs. Kali Linux, running the Zabbix monitoring server, is connected to the same network segment to monitor all infrastructure devices.



### 3. IP Addressing and Monitoring Table

The table below lists the devices in scope for monitoring along with their assigned IP addresses and monitoring method.

| Device                     | IP Address   | Monitoring Method    |
|----------------------------|--------------|----------------------|
| Kali Linux / Zabbix Server | 192.168.1.10 | Zabbix Server (Host) |
| ISP-R1                     | 10.0.0.1:1   | ICMP + SNMP          |
| Home-R1                    | 192.168.1.1  | ICMP + SNMP          |
| SW1                        | 192.168.1.2  | ICMP + SNMP          |
| PC1                        | DHCP IP      | Optional             |
| PC2                        | DHCP IP      | Optional             |
| PC3                        | DHCP IP      | Optional             |

### 4. Zabbix Server Configuration

Zabbix Server and its required services (Zabbix server process and web server) were verified as running on Kali Linux prior to configuring monitoring. The Zabbix frontend was accessed via a web browser to confirm the installation.

```

Kali 2025 x64 Customized x Eve-ng x
Jul 28 04:10
kali@kali: ~

● zabbix-server.service - Zabbix Server (MySQL/MariaDB)
   Loaded: loaded (/usr/lib/systemd/system/zabbix-server.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-07-28 02:03:26 CDT; 2h 6min ago
   Invocation: 44cf09a2a74d4c22a25b997c91722cc9
     Docs: man:zabbix_server
    Main PID: 3254 (zabbix_server)
      Tasks: 64 (limit: 4606)
    Memory: 105.1M (peak: 106.5M)
       CPU: 40.072s
    CGroup: /system.slice/zabbix-server.service
           └─3254 /usr/sbin/zabbix_server --foreground
              └─3258 "/usr/sbin/zabbix_server: ha manager"
                 └─3259 "/usr/sbin/zabbix_server: service manager #1 [processed 0 events, updated 0 event tags, deleted 0 problems, synced 0 service updates, idle 10 sec]"
                    └─3260 "/usr/sbin/zabbix_server: configuration synchronizer [syncd configuration in 0.026876 sec, idle 10 sec]"
                       └─3261 "/usr/sbin/zabbix_server: alert manager #1 [sent 0, failed 0 alerts, idle 5.045429 sec during 5.045711 sec]"
                          └─3262 "/usr/sbin/zabbix_server: alerter #1 started"
                             └─3263 "/usr/sbin/zabbix_server: alerter #2 started"
                                └─3264 "/usr/sbin/zabbix_server: alerter #3 started"
                                   └─3265 "/usr/sbin/zabbix_server: preprocessing manager #1 [queued 89, processed 97 values, idle 5.297232 sec during 5.298739 sec]"
                                      └─3266 "/usr/sbin/zabbix_server: lld manager #1 [processed 0 LLD rules, idle 5.870089sec during 5.870430 sec]"
                                         └─3267 "/usr/sbin/zabbix_server: lld worker #1 [processed 2 LLD rules, idle 221.783864 sec during 221.798973 sec]"
                                            └─3268 "/usr/sbin/zabbix_server: lld worker #2 [processed 1 LLD rules, idle 107.969434 sec during 107.975177 sec]"
                                               └─3269 "/usr/sbin/zabbix_server: housekeeper [deleted 3 hist/trends, 1686 items/triggers, 0 events, 0 sessions, 0 alarms, 0 audit items, 0 auto..."
                                                  └─3270 "/usr/sbin/zabbix_server: timer #1 [updated 0 hosts, suppressed 0 events in 0.001166 sec, idle 59 sec]"
                                                     └─3271 "/usr/sbin/zabbix_server: http poller #1 [got 0 values in 0.000057 sec, idle 5 sec]"
                                                        └─3275 "/usr/sbin/zabbix_server: browser poller #1 [got 0 values in 0.000055 sec, idle 5 sec]"
                                                           └─3276 "/usr/sbin/zabbix_server: discovery manager #1 [processing 0 rules, 0 unsaved checks]"
                                                              └─3277 "/usr/sbin/zabbix_server: history synchronizer #1 [processed 0 values, 0 triggers in 0.000020 sec, idle 1 sec]"
                                                                 └─3278 "/usr/sbin/zabbix_server: history synchronizer #2 [processed 0 values, 0 triggers in 0.000022 sec, idle 1 sec]"
                                                                    └─3279 "/usr/sbin/zabbix_server: history synchronizer #3 [processed 0 values, 0 triggers in 0.000032 sec, idle 1 sec]"
                                                                       └─3280 "/usr/sbin/zabbix_server: history synchronizer #4 [processed 0 values, 0 triggers in 0.000140 sec, idle 1 sec]"
                                                                          └─3281 "/usr/sbin/zabbix_server: escalator #1 [processed 0 escalations in 0.001636 sec, idle 3 sec]"
                                                                             └─3282 "/usr/sbin/zabbix_server: proxy poller #1 [exchanged data with 0 proxies in 0.000047 sec, idle 5 sec]"
                                                                                └─3283 "/usr/sbin/zabbix_server: self-monitoring [processed data in 0.000190 sec, idle 1 sec]"
                                       └─3284 "/usr/sbin/zabbix_server: task manager [processed 0 task(s) in 0.000724 sec, idle 5 sec]"
                                          └─3285 "/usr/sbin/zabbix_server: poller #1 [got 0 values in 0.000068 sec, idle 5 sec]"
                                             └─3286 "/usr/sbin/zabbix_server: poller #2 [got 0 values in 0.000027 sec, idle 5 sec]"
                                                └─3287 "/usr/sbin/zabbix_server: poller #3 [got 0 values in 0.000025 sec, idle 5 sec]"
                                                   └─3288 "/usr/sbin/zabbix_server: poller #4 [got 0 values in 0.000068 sec, idle 5 sec]"

```

The screenshot displays the Zabbix Server Dashboard. The top navigation bar includes links for Dashboards, Monitoring, Services, Inventory, Reports, Data collection, Alerts, Users, and Administration. The main content area shows a global view with several widgets: 'Top hosts by CPU utilization' with a bar chart, 'System information' table, 'Host availability' status indicators, 'Problems by severity' bar chart, and a 'Geomap' showing a location pin. A large digital clock displays '04:10 Chicago'. The browser's address bar shows the URL '192.168.1.10/zabbix/zabbix.php?action=dashboard.view&dashboardid=1'. A tooltip for 'Zabbix Technical Support' is visible at the bottom left, and a URL 'https://www.zabbix.com/support' is shown below the browser window.

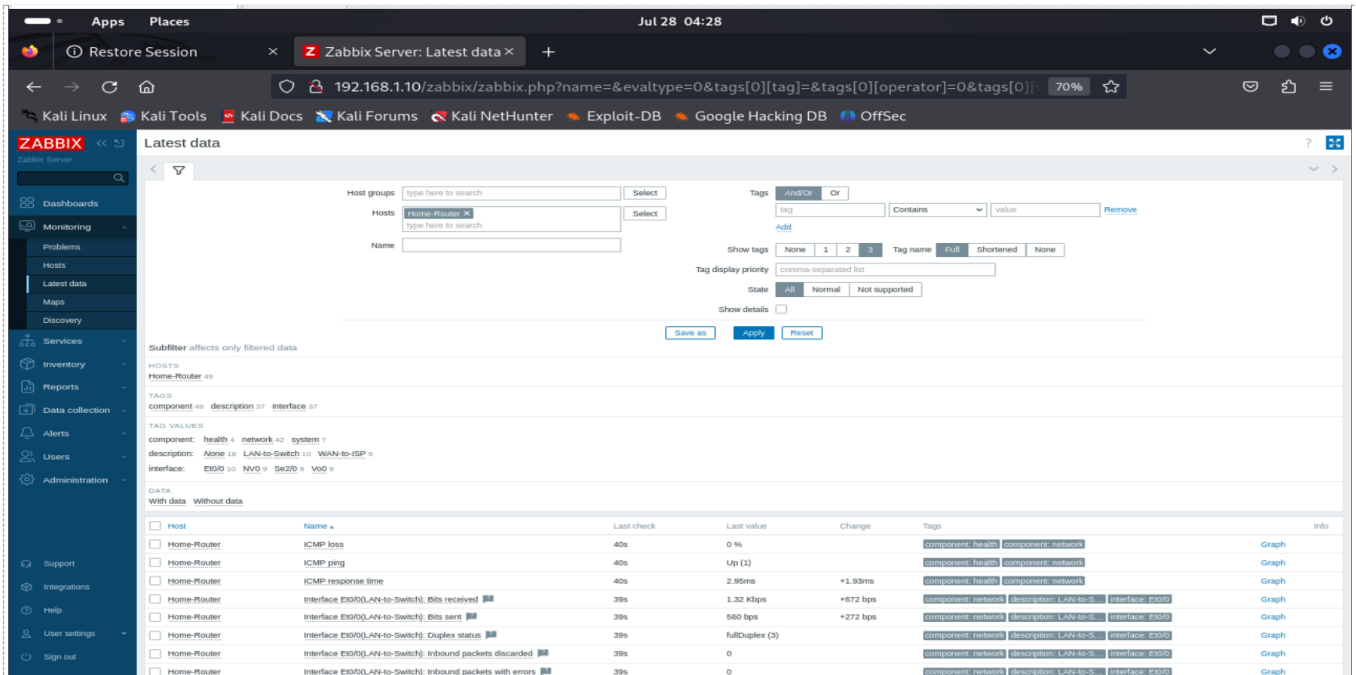
## 5. Host Configuration

Home-R1, ISP-R1, and SW1 were added to Zabbix as monitored hosts, each assigned to the "Network Devices" host group and configured with their respective management IP addresses.

The screenshot shows the Zabbix Hosts configuration page. The top navigation bar includes links for Hosts, Templates, Maintenance, Event correlation, Discovery, Alerts, Users, and Administration. The main content area features a form for adding new hosts with fields for Name, DNS, IP, and Port, and a 'Status' dropdown. Below the form is a table of existing hosts. The table has columns for Name, Items, Triggers, Graphs, Discovery, Web, Interface, Proxy, Templates, Status, Availability, Agent encryption, Info, and Tags. The hosts listed are Home-Router, ISP-Router, Switch, and Zabbix server. A tooltip for 'Zabbix Integrations' is visible at the bottom left, and a URL 'https://www.zabbix.com/integrations' is shown below the browser window.

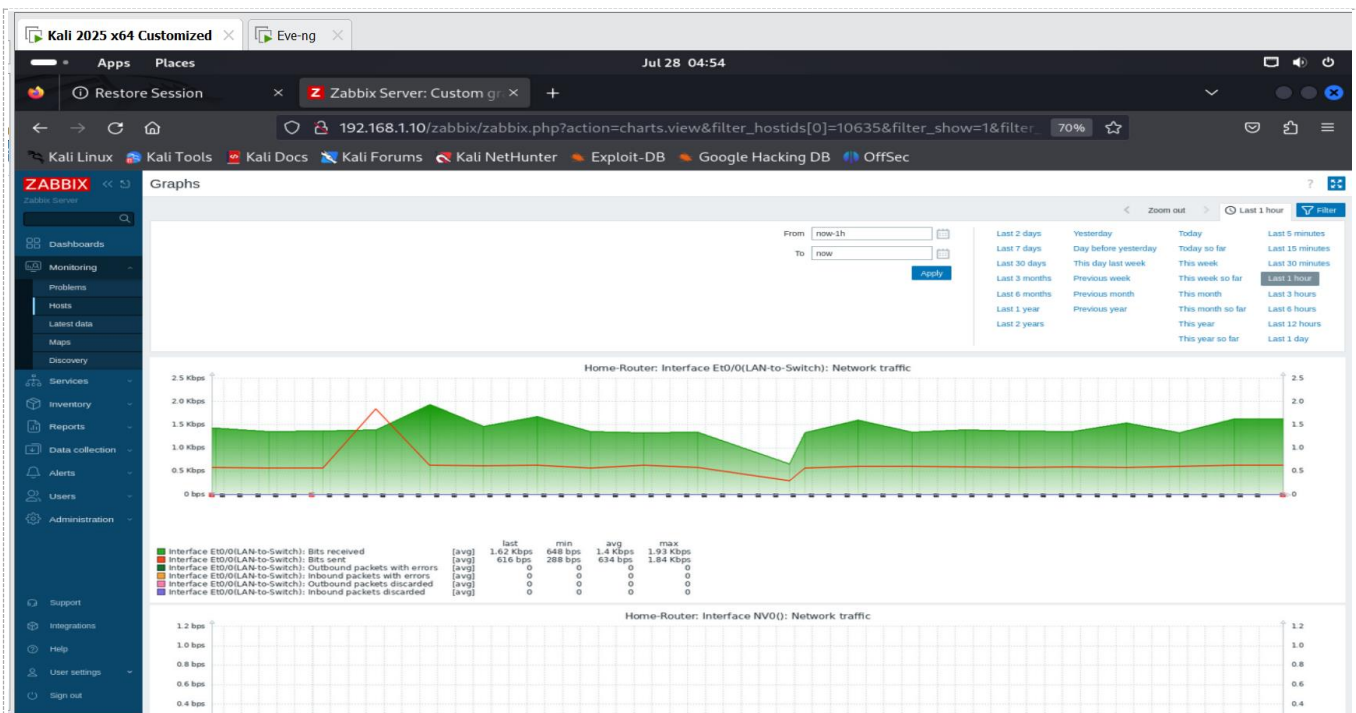
## 6. ICMP Monitoring

Each host was configured with ICMP checks to monitor basic reachability, latency, and packet loss. This provides real-time visibility into whether each device is up or down.



## 7. SNMP Monitoring

In addition to ICMP, SNMP monitoring was configured on the network devices to collect deeper operational metrics, including interface status, interface traffic, CPU utilization, memory usage, and interface errors.



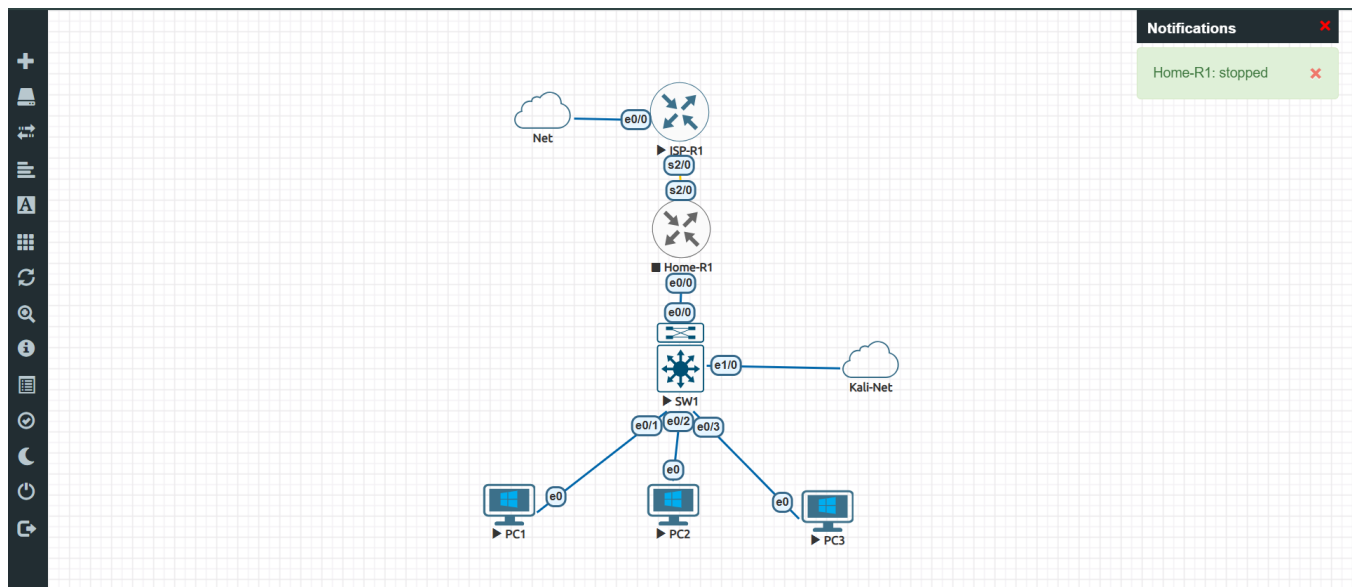
## 8. Trigger and Alert Configuration

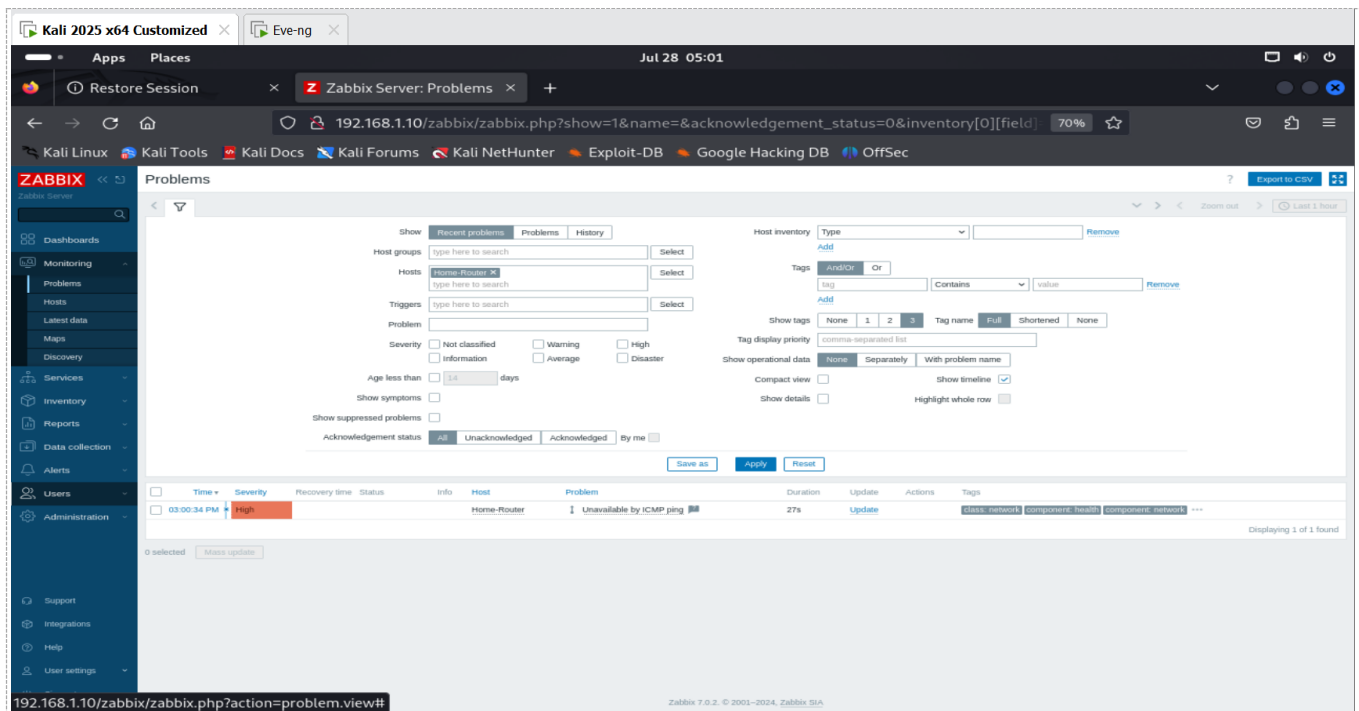
A trigger was created for each host to raise a Problem in Zabbix when the device becomes unreachable (ICMP check fails). This ensures that any outage is immediately flagged for NOC investigation.

| Severity | Value | Name                                                                        | Operational data                                            | Expression                                                                                                                                                                                                                                                                                                                                                          | Status  | Info | Tags                |
|----------|-------|-----------------------------------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------|---------------------|
| High     | OK    | Network Generic Device by SNMP: Unavailable by ICMP ping                    |                                                             | <code>max([Home-Router/icmping.#3])=0</code>                                                                                                                                                                                                                                                                                                                        | Enabled |      | scope: availability |
| Average  | OK    | Network interfaces discovery: Interface Se2/0 (WAN-to-ISP): Link down       | Current state: {ITEM.LASTVALUE1}                            | Problem: <code>{BIFCONTROL."Se20"}=1</code> and <code>last([Home-Router/if status[OperStatus.9]=2)</code><br>Recovery: <code>last([Home-Router/if status[OperStatus.9]=2)</code> or <code>{BIFCONTROL."Se20"}=0</code>                                                                                                                                              | Enabled |      | scope: availability |
| Average  | OK    | Network interfaces discovery: Interface E5/0 (LAN-to-Switch): Link down     | Current state: {ITEM.LASTVALUE1}                            | Problem: <code>{BIFCONTROL."E50"}=1</code> and <code>last([Home-Router/if status[OperStatus.1]=2)</code><br>Recovery: <code>last([Home-Router/if status[OperStatus.1]=2)</code> or <code>{BIFCONTROL."E50"}=0</code>                                                                                                                                                | Enabled |      | scope: availability |
| Average  | OK    | Network interfaces discovery: Interface NV0/1                               | Current state: {ITEM.LASTVALUE1}                            | Problem: <code>{BIFCONTROL."NV0"}=1</code> and <code>last([Home-Router/if status[OperStatus.13]=2)</code><br>Recovery: <code>last([Home-Router/if status[OperStatus.13]=2)</code> or <code>{BIFCONTROL."NV0"}=0</code>                                                                                                                                              | Enabled |      | scope: availability |
| Average  | OK    | Network interfaces discovery: Interface Vo0/1                               | Current state: {ITEM.LASTVALUE1}                            | Problem: <code>{BIFCONTROL."Vo0"}=1</code> and <code>last([Home-Router/if status[OperStatus.13]=2)</code><br>Recovery: <code>last([Home-Router/if status[OperStatus.13]=2)</code> or <code>{BIFCONTROL."Vo0"}=0</code>                                                                                                                                              | Enabled |      | scope: availability |
| Warning  | OK    | Network interfaces discovery: Interface Se2/0 (WAN-to-ISP): High error rate | errors in: {ITEM.LASTVALUE1}, errors out: {ITEM.LASTVALUE2} | Problem: <code>min([Home-Router/if in.errors[InErrors.9.5m]&gt;{BIFERRORS.WARN."Se20"} or min([Home-Router/if out.errors[OutErrors.9.5m]&gt;{BIFERRORS.WARN."Se20"}]</code><br>Recovery: <code>max([Home-Router/if in.errors[InErrors.9.5m]&lt;{BIFERRORS.WARN."Se20"}=0.8 and max([Home-Router/if out.errors[OutErrors.9.5m]&lt;{BIFERRORS.WARN."Se20"}=0.8</code> | Enabled |      | scope: performance  |
| OK       | OK    | Network interfaces discovery: Interface Se2/0 (WAN-to-ISP): Link down       | Current state: {ITEM.LASTVALUE1}                            | <code>max([Home-Router/if status[OperStatus.9]=2)</code> or <code>{BIFCONTROL."Se20"}=0</code>                                                                                                                                                                                                                                                                      | Enabled |      | scope: availability |

## 9. Simulated Incident: Home-R1 Failure

With the environment in a healthy state (ISP-R1, Home-R1, and SW1 all reachable), a controlled failure was introduced by disconnecting/shutting down the Home-R1 interface in EVE-NG to simulate a real outage.





## 10. Incident Respond

The table below documents the full incident lifecycle for INC-001 in a format consistent with standard NOC/SOC incident reporting practices.

|                   |                                                                                                                                                                                                                                      |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Incident ID       | INC-001                                                                                                                                                                                                                              |
| Device            | Home-R1                                                                                                                                                                                                                              |
| Monitoring System | Zabbix                                                                                                                                                                                                                               |
| Detection Method  | ICMP Monitoring                                                                                                                                                                                                                      |
| Problem           | Home-R1 became unreachable.                                                                                                                                                                                                          |
| Detection Time    | 2026-07-28 03:00:34 PM                                                                                                                                                                                                               |
| Investigation     | Zabbix generated an alert indicating that Home-R1 was unreachable. Ping testing confirmed that Home-R1 was not responding. ISP-R1 and SW1 remained reachable, indicating the fault was isolated to Home-R1 or its direct connection. |
| Root Cause        | Home-R1 Device powered off                                                                                                                                                                                                           |
| Resolution        | Restored Home-R1 connectivity.                                                                                                                                                                                                       |
| Verification      | Home-R1 responded successfully to ICMP ping. Zabbix returned the device to normal status.                                                                                                                                            |
| Status            | Resolved                                                                                                                                                                                                                             |



```

(kali@kali)-[~]
$ ping -c 4 192.168.1.2
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.
64 bytes from 192.168.1.2: icmp_seq=1 ttl=255 time=1.01 ms
64 bytes from 192.168.1.2: icmp_seq=2 ttl=255 time=1.60 ms
64 bytes from 192.168.1.2: icmp_seq=3 ttl=255 time=1.58 ms
64 bytes from 192.168.1.2: icmp_seq=4 ttl=255 time=2.08 ms

--- 192.168.1.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3020ms
rtt min/avg/max/mdev = 1.006/1.564/2.076/0.379 ms

(kali@kali)-[~]
$ ping -c 4 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
From 192.168.1.10 icmp_seq=1 Destination Host Unreachable
From 192.168.1.10 icmp_seq=2 Destination Host Unreachable
From 192.168.1.10 icmp_seq=3 Destination Host Unreachable
From 192.168.1.10 icmp_seq=4 Destination Host Unreachable

--- 10.0.0.1 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3031ms
pipe 2

(kali@kali)-[~]
$

```

---

```

(kali@kali)-[~]
$ ping -c 4 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=255 time=1.89 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=255 time=2.52 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=255 time=3.52 ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=255 time=1.65 ms

--- 192.168.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3028ms
rtt min/avg/max/mdev = 1.645/2.394/3.517/0.723 ms

(kali@kali)-[~]
$ ping -c 4 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=254 time=28.2 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=254 time=33.0 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=254 time=34.0 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=254 time=49.0 ms

--- 10.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3027ms
rtt min/avg/max/mdev = 28.167/36.061/49.046/7.815 ms

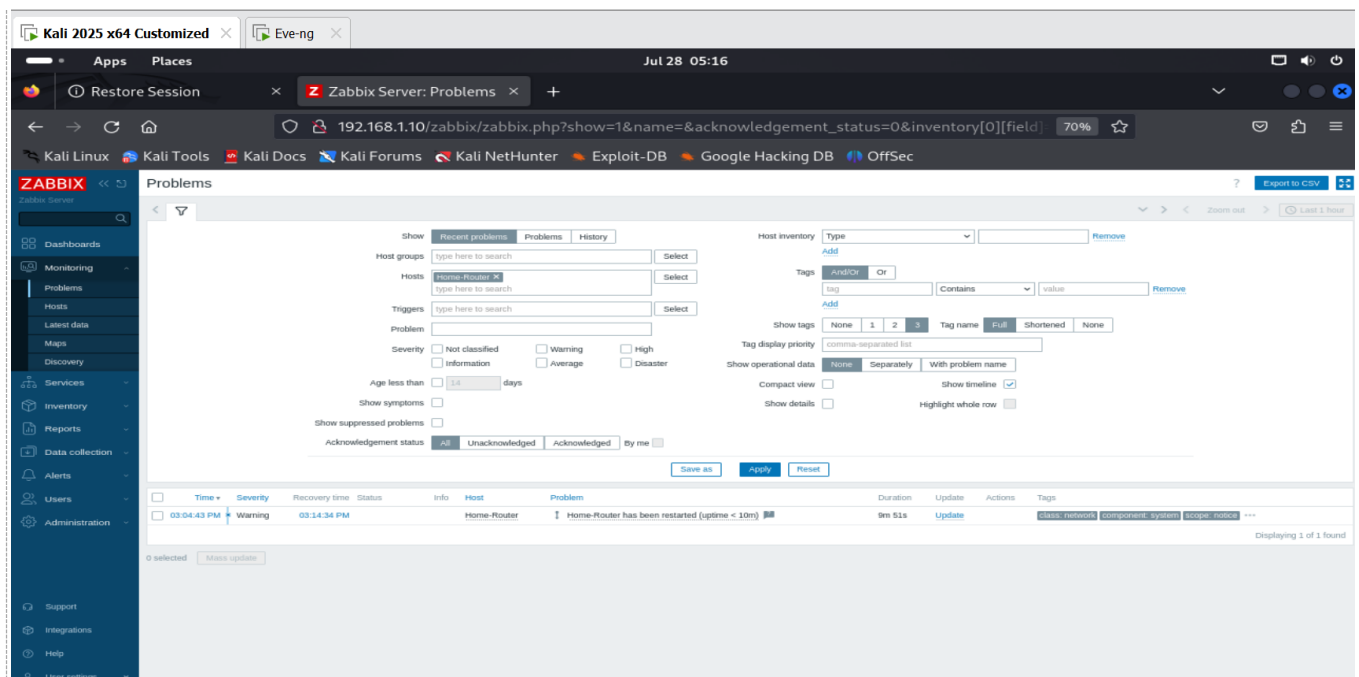
(kali@kali)-[~]
$ ping -c 4 192.168.1.2
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.
64 bytes from 192.168.1.2: icmp_seq=1 ttl=255 time=1.57 ms
64 bytes from 192.168.1.2: icmp_seq=2 ttl=255 time=1.83 ms
64 bytes from 192.168.1.2: icmp_seq=3 ttl=255 time=1.60 ms
64 bytes from 192.168.1.2: icmp_seq=4 ttl=255 time=1.71 ms

--- 192.168.1.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3023ms
rtt min/avg/max/mdev = 1.534/1.605/1.711/0.066 ms

(kali@kali)-[~]
$

```





## 11. Project Deliverables Summary

- EVE-NG topology with ISP-R1, Home-R1, SW1, PC1-PC3, and Kali Linux/Zabbix server
- Zabbix dashboard showing all monitored devices in a healthy state
- ICMP monitoring: availability, latency, packet loss
- SNMP monitoring: interface status, traffic, CPU, memory, errors
- Configured trigger for device unreachability
- Simulated Home-R1 outage
- Full NOC investigation and root cause analysis
- Documented resolution and verification
- Completed incident report (INC-001)

## 12. Conclusion

This project simulated a real-world NOC monitoring and incident response workflow, from initial detection through resolution and documentation. It demonstrates practical experience with Zabbix configuration, network monitoring (ICMP/SNMP), and structured incident handling core skills required in a NOC or SOC analyst role.